Weekly Report

Triage Classifier: the GNN Gate and the LLM Arm

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This Week's Changes

- **Direction:** replace the balanced hypothesis screen (median 1.1 s per alarm, 229 refits per round) by a classifier that reads only the balanced WLS solve. Two arms: a graph neural network, and the LLM itself.
- **GNN: 100%** recall with **0.9%** unneeded phasor requests in 2 ms; the screen gets 98.5% and 12.0%. No simulator-background effect.
- **LLM as the classifier:** six fine-tunes over three prompt forms; best greedy recall 83%. Trees on the same prompt fields reach 98.5%. Closed for IEEE 14.
- **GNN-gated profile:** the rule expert solves **159/160** development roots in 4.1 min, the same roots as with the screen in 10 min.
- **DAgger cell with the GNN gate:** round 1 expert 159, BC0 153, **R1 159** of 160 (screen cell: 160, 152, 158). Round 2 is in evaluation.
- **Trace review:** no family leak found; four contract changes cut tool calls by 9.4% at equal success.

Why Replace the Screen

Cost. After every alarm the screen refits the operator's balanced model under each single cause.

Class	Refits per round
Meter	6
Parameter (R, X, both)	60
Outage	19
HIF (16 lines $\times$ 9 positions)	144
One round	**229**

Median **1.1 s** per alarm against 0.05 s for the WLS itself, and the count grows with the branch count.

Source: S1 (sections 1–2), S9 (screen cost, October 2).

The LLM's role. The agent read the screen's verdict in its prompt, so the LLM never classified anything from the WLS result.

Decisions of October 4

G1	The classifier's thresholded score gates a request for phase-resolved measurements.
T1	Labels come from the truth: a request is needed when it holds an HIF, an unbalance or a harmonic source.
S1	Start with an offline benchmark on IEEE 14; IEEE 57 and 118 later.

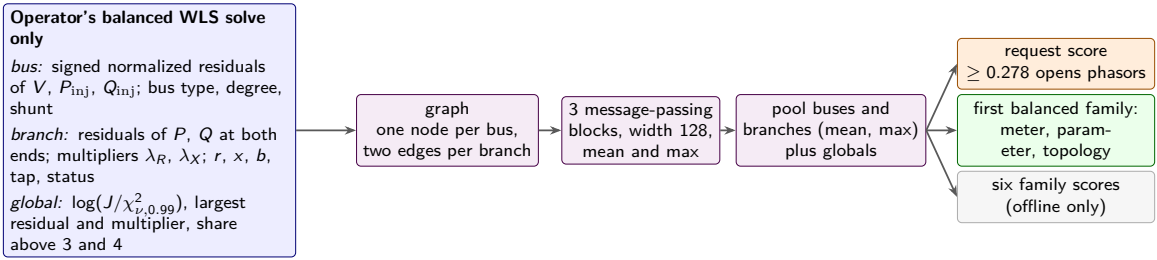
Two arms: a GNN whose score is the gate, and the LLM deciding by itself.

What the Classifier Decides

- **Two decisions.** Request PMU phasors or not, and which balanced family to try first: meter, parameter or topology. Six family scores are a diagnostic only.
- **Labels (T1).** A request is needed when the truth holds an HIF, an unbalance or a harmonic source. On a mixed root, the first family is the one whose removal lowers J most.
- **Not the classifier's job.** Locating the HIF: the zero-sequence line differential on the acquired phasors does that.
- **Fallbacks that need no screen.** Phasors open when every balanced correction on the state was rejected by verification, or none was offered while the alarm stands. Spectra follow phasors that came back balanced.
- **Operating point.** Every learned score is thresholded on the calibration split at the screen rule's recall (99.3%), so the tables compare unneeded requests at equal recall.

Source: S1 (sections 2, 3 and 5).

GNN Triage Model



- 706,506 parameters, five seeds averaged; no weight depends on the network size. 2 ms per alarm on a CPU; the export is 6.9 MB.
- **Residual-only view:** no measured or fitted value, no estimated state, no sigma. Those carried the September family cues and tie a model to one operating range.
- Injection residuals at zero-injection buses are zeroed on every row: only the topology roots treat those buses as exact.

Source: S5 (`features.py`, `model.py`, `runtime.json`).

Offline Benchmark on IEEE 14

- **Rows:** alarmed states of the step-4 study, split by physical parent: 2,277 train (with truth-corrected children), 458 calibration, **463 test**. Of the test roots, 130 need phasors and 333 need none, including alarmed healthy windows and flow-meter pairs that mimic an HIF.
- **References:** the screen rule, and gradient-boosted trees on five inputs: the screen's outputs, 45 WLS-only features, the prompt's WLS view (five largest residuals and multipliers), ten or twenty signed residuals, and every residual and multiplier as one flat vector.
- **Cue checks:**
  - *simulator background:* 1,311 probe rows, each a healthy window plus one bad meter, on an OpenDSS or an OPF background; no probe row needs phasors;
  - *location:* three folds over the 26 waveform locations, each held out of training;
  - *operating point:* a "values" view that adds measured and fitted values.
- Brackets in the tables are 95% parent-bootstrap intervals.

Source: S1 (section 5), S2.

Request Decision: GNN Against the Screen

Test split: 130 roots need phasors, 333 need none; learned scores at the screen's calibration recall

Classifier	Recall	Unneeded requests	Time per alarm
Screen rule (reference)	98.5% [95.9, 100]	12.0% [8.7, 16.0]	1.1 s
Trees, screen outputs	96.9% [93.8, 99.3]	1.2% [0.3, 2.4]	screen + 7 ms
Trees, 45 WLS-only features	98.5% [96.2, 100]	10.5% [7.4, 13.9]	4 ms
Trees, the prompt's WLS view	100%	12.9% [9.5, 16.6]	4 ms
Trees, ten signed residuals	98.5% [96.3, 100]	5.4% [3.0, 8.3]	4 ms
Trees, flat full vector	97.7% [94.7, 100]	10.2% [7.0, 13.9]	4 ms
GNN, residual-only	**100%**	**0.9% [0.0, 2.1]**	**2 ms**
GNN, with values	100%	0.9% [0.0, 2.1]	2 ms

- The GNN requests on all 130 roots that need phasors and on 3 of the 333 that do not (two single-meter roots, one multi-meter root). Test AUC is 99.88–99.98% by seed.
- The flat full vector does no better than the 45 hand-built features: the gain comes from the graph structure, not from more numbers. Values add nothing here and transfer worse (slide 9).

Source: S2 (`report.md`), S1 (section 6).

Does the GNN Read Cues?

Probes: healthy window plus one bad meter

		GNN requests	
Bad meter	Screen	OpenDSS	OPF
Power, 10–20σ	0%	1.7%	1.3%
Voltage, 10–20σ	97%	11.8%	9.6%
Voltage, 4.5–9σ	94%	11.7%	9.8%

Recall at held-out waveform locations

Family	Unseen	Seen
All	97.0% (260/268)	99.6%
HIF	97.3% (107/110)	99.0%
Unbalance	91.7% (55/60)	100%
Harmonic	100% (98/98)	100%

Source: S2 (probe tables), S3 (location study), S1 (section 6).

- **No background effect.** The backgrounds differ by 0.4 to 2.3 points; every interval includes zero.
- **No size cue.** Small voltage-meter errors, below the training range, get the rate of large ones. Trees on the prompt view request on 73–95% of them.
- The screen requests on almost every bad voltage meter by design: most of its 12%.
- **Unseen locations** cost 3 points of recall, 8 for unbalance. The 8 misses are confident (scores ≤ 0.06).
- **Flow-meter pairs.** The GNN's HIF head flags none of 16 same-sign pairs; the screen flags 37.5%.

First Balanced Family, and a First Look at IEEE 57

First balanced family (rows that need no phasors)

Classifier	True family	Oracle order on mixed roots
Screen, first winner	93.1%	99.0%
Trees, screen outputs	98.7%	99.0%
GNN, residual-only	96.5%	92.7%
Trees, the prompt's view	96.9%	94.8%

IEEE 57 zero-shot (184 alarmed rows)

Classifier	Recall	Requests on healthy rows
Screen rule	63.9%	3/26
GNN, residual-only	50.0%	0/26
GNN, with values	31.0%	0/26

- The GNN names a true balanced family more often than the screen's first winner, but follows the oracle order on mixed roots less often.
- "No retraining across networks" is not supported yet: with the IEEE-14 weights and threshold, half the recall on IEEE 57 (AUC 84.2%). The residual view transfers better than the values view, which supports the feature choice.
- The IEEE 57 rows hold 112 HIF, 46 unbalance and 26 healthy windows and no balanced fault yet. A fair test needs training on more than one network.

Source: S2 (`transfer_look_ieee57.md`), S1 (sections 6–7).

LLM Arm: Can the LLM Classify by Itself?

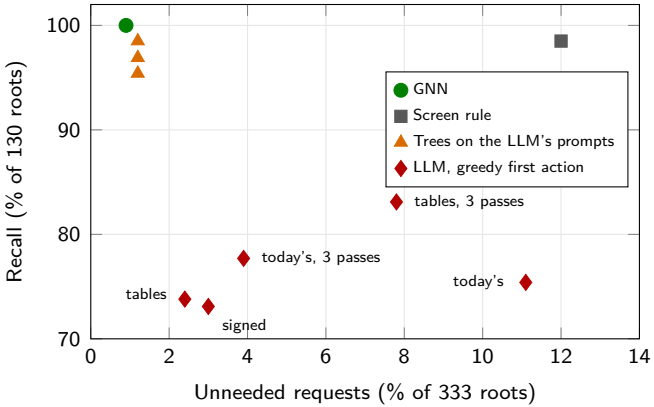
- **Rows:** the agent's decision right after the opening WLS, rendered as in the DAgger pipeline with no screen (4,200–5,300 tokens); 2,035 train, 242 validation, the same 463 test roots.
- **Target:** request phasors when the truth needs them; otherwise open the context of the first balanced family.
- **Recipe:** the pipeline's BC0 fine-tune: Gemma 4 12B in 4 bits, rank-16 LoRA, one or three passes, best validation loss kept.
- **Two readings:** the greedy first action, and the probability of the request token thresholded like every other score.
- **Same-prompt control:** trees on fields parsed from the same rendered prompts, with the same rows and target. A gap to the control is a gap of the learner, not of the prompt.

Prompt form	WLS summary shown to the LLM
Today's	up to five residual magnitudes of 3σ or more, five branch multipliers
Signed residuals	ten residuals with signs requested; the model view's list cap keeps eight
Bus and branch tables	the alarm's neighbourhood as tables; about 1,200 more tokens

Source: S1 (section 5), S4.

LLM Arm: Request Decision

Same 463 test roots; each point is one classifier read at its own decision



- Every one-pass LLM: **73–75%** recall, whatever the prompt.
- Best LLM: **83.1%** at 7.8% unneeded (tables, three passes at half the learning rate).
- Trees on the same prompt fields: **95–98.5%** at 1.2%.
- As a ranking the LLM reaches AUC 97.1% at best (GNN 99.96%). At the screen's recall it requests on 46–85% of the roots that need none.

Source: S2 (`report.md`), S4. Every reading: appendix.

LLM Arm: What the Fine-Tune Learned

Requests on the test roots that need phasors, by family

		LLM, greedy						
Family	Roots	today's	signed	today's $\times 3$	tables	tables $\times 3$	Trees	GNN
Harmonic	50	50	50	50	50	50	50	50
Unbalance	30	28	28	26	28	28	28–29	30
HIF	25	17	17	18	17	21	24–25	25
HIF + bad meter	25	3	0	7	1	9	22–24	25

- **One cue.** A depth-3 tree on the prompt fields reproduces 94–98% of the LLM's request decisions. Its first split: the largest residual is a voltage magnitude; its second: that residual is not dwarfed by the largest branch multiplier.
- A voltage residual leads the list on 84 of the 130 roots that need phasors: every harmonic root, 29 of 30 unbalance, 5 of 25 HIF and no HIF behind a bad meter. With today's prompt the LLM requests on 100% of the 10–20σ voltage-meter probes and on 10% of the bad power meters.
- The tables carry the information, since trees on them reach 98.5%; the LLM's first action does not use it.

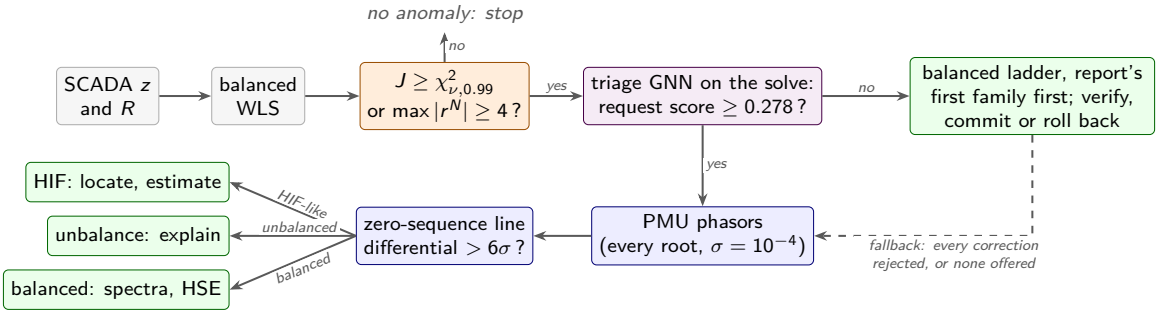
Source: S1 (section 6), S4. Trees: largest class on the three prompt forms.

LLM Arm: Verdict for IEEE 14

- **Closed as a classifier.** With this recipe the LLM is not a triage classifier, so no DAgger cell uses the LLM as the gate.
- **More passes trade one error for the other.** Today's prompt, three passes: unneeded requests 11.1% $\rightarrow$ 3.9%, recall 75% $\rightarrow$ 78%. Tables at half the learning rate: recall 83%, unneeded 7.8%.
- **The recipe is fragile here.** One three-pass run diverged at the peak learning rate and ranked worse than chance (AUC 28.8%).
- **The prompt is not the limit.** Trees on the same fields reach 95–98.5% recall at 1.2% unneeded.
- **What stays meaningful:** the agent acting on the GNN's report, against the agent acting on the screen's report.
- **Not tried:** more training roots, a larger model, a reasoning trace before the decision.

Source: S1 (sections 7–8), S4.

The GNN-Gated Evidence Profile



- The report sits on the WLS ledger: request score, threshold, admission and the first balanced family. Nothing is added to the fault signatures.
- Same data contract as the screen profile: every root carries phasors and spectra. A classifier failure admits nothing, and the solve stands.

Source: S1 (section 10), S5, S10 (`evidence_profile.py`, `actions.py`).

Expert End to End: GNN Gate Against the Screen

Rule expert on the 160 development roots, 40 actions, one CPU

Profile	Expert	Success	Phasors (unneeded)	Spectra (unneeded)	Mean steps	Wall time
GNN-gated	baseline	159/160	59 (3)	18 (2)	8.0	**4.1 min**
Screen	baseline	159/160	59 (3)	17 (1)	8.6	10.0 min
Screen	ledger and ranker	160/160	58 (2)	16 (0)	7.9	9.4 min

- The GNN's first report requests on all 40 roots that need phasors and on 1 of the 104 that need none (a multi-meter root at score 0.31).
- With the same expert, both profiles solve the same 159 roots and fail the same one. The GNN profile takes 2.4 times less wall time.
- The failing root is a parameter error whose true line ranks second among the multipliers (ratio 1.02). The screen's ledger names the right line; the baseline expert commits the wrong one. That is localization, not the request decision.

Source: S1 (section 10), S6. "Unneeded" leaves out the 16 harmonic roots, where phasors come before the spectra for every teacher.

DAgger Cell with the GNN Gate

Gemma 4 12B, GNN-gated profile, baseline expert; the 160 development roots of the Oct 1 screen cell

Stage	Result
D0 aggregate	525 of 548 roots, 4,376 rows
BC0	826 steps; validation loss 0.0048
Round 1	122 roots, 572 rows (label yield 86%); R1 trained on 1,144 rows
Round 2	122 roots, 537 rows; R2 trained; evaluation restarted after a preemption

Round 1	Expert	BC0	R1
Successes	159	153	**159**
Invalid-action episodes	0	10	3
False-commit episodes	0	1	1
Loop episodes	1	9	0
Mean steps	8.01	9.16	8.02
Screen cell, Oct 1	160	152	158

- R1 matches the expert family by family; both miss only the misranked parameter root.
- BC0 loses 5 of the 8 HIF roots behind a bad meter, one topology root and the misranked root.
- The screen cell's round 2 fell to 156, every loss after balanced phasors on a voltage-meter suspicion.

Source: S7 (`out/r1/round_summary.json`, round-2 receipts), S8.

Trace Review: Leaks and Redundant Calls

Family leaks: none found

- Every root carries phasors and spectra at the same sigma; no request answers by availability.
- The family-independent prompt fields are identical across the 13 families.
- Fault signatures come only from the agent's own diagnostics.
- The probes show no simulator-background effect.

HIF, harmonic, unbalance and healthy traces were already at the minimum the gates allow.

Source: S1 (section 12), S10 (commits 18c15f7, 11b7ad5).

Four changes for the next cell

1. **No family scores in the report.** Harmonic and unbalance differ from a bad voltage meter only through simulator artifacts.
2. **No phasor re-acquisition** on a corrected child once they were tested.
3. **One confirmation handoff** when a committed correction leaves the WLS quiet. Finalizing stays forbidden: an edit that only silences the alarm must not close the case.
4. **Conditioned WLS inside the HIF estimate**, run by the environment, outside the agent's action budget.

Trimmed Contract: Old and New Traces

Root	Before	Now
HIF	WLS → phasors → NLM → HIF estimate → **WLS** → finalize	WLS → phasors → NLM → HIF estimate (conditioned WLS inside) → finalize
Meter, parameter or topology	WLS → context → correct → WLS → commit → **context** → handoff	WLS → context → correct → WLS → commit → confirmation handoff
HIF + bad meter	. . . HIF estimate → **WLS** → . . . commit → **context** → handoff	. . . HIF estimate → . . . commit → confirmation handoff

160 development roots	GNN-gated, baseline expert		Screen, ledger and ranker	
	Before	Now	Before	Now
Successes	159	159	160	160
Tool calls	1,282	**1,161**	1,256	**1,137**
Calls per HIF root	6	5	6	5
Calls per HIF + bad meter root	11.4	9.1	11.0	9.1
Calls per single balanced fault	7	6	7	6

Source: S6 (expert end-to-end runs before and after), S1 (section 12). Harmonic, unbalance and healthy roots are unchanged at 6, 4 and 2 calls.

Open Problems

- **Misranked parameter root.** The baseline expert commits the wrong line when the true one ranks second. The ledger teacher solves it in 6 calls; the GNN profile needs that ranking without the screen's refits.
- **Classifier false admissions.** A multi-meter root at the first WLS (score 0.31) and a meter-plus-topology child after its fix (0.40) each cost phasor and spectra calls.
- **No admission and nothing to try.** The episode ends in a handoff rather than a phasor request: the fallback needs a rejected correction, or a context that offered nothing.
- **One network.** The GNN is trained on IEEE 14 only, and its zero-shot IEEE 57 recall is 50%.
- **Corpus format.** Topology roots are the only ones on the node-breaker operator model. The features neutralize it; one operator model for all roots would remove it.

Source: S1 (sections 7, 8, 10 and 12), S6.

Next Experiments

1. **Finish round 2.** R2 against R1, and against the screen cell's R2 (156), on the same 160 roots.
2. **Next cell on the trimmed contract.** The settings are ready; it needs about a day of GPU time.
3. **A multi-network GNN.** IEEE 57 and 118 roots of every family, then leave-one-network-out.
4. **Balanced localization without the screen.** Carry the ledger's candidate ranking into the parameter route.
5. **One operator model for all roots**, to remove the last corpus-format cue.

Appendix: Every Request Reading on the Test Split

Classifier	Reading	AUC	Recall	Unneeded	Balanced pick true
LLM, today's, 1 pass	greedy	–	75.4%	11.1%	66.0%
LLM, signed, 1 pass	greedy	–	73.1%	3.0%	82.1%
LLM, today's, 3 passes	greedy	–	77.7%	3.9%	80.5%
LLM, tables, 1 pass	greedy	–	73.8%	2.4%	83.3%
LLM, tables, 3 passes, half rate	greedy	–	83.1%	7.8%	76.4%
LLM, today's, 1 pass	probability	91.5%	100%	85.3%	77.4%
LLM, signed, 1 pass	probability	95.6%	99.2%	75.7%	85.2%
LLM, today's, 3 passes	probability	94.7%	99.2%	55.3%	84.3%
LLM, tables, 1 pass	probability	97.1%	100%	45.6%	85.8%
LLM, tables, 3 passes, half rate	probability	95.7%	100%	49.2%	85.2%
Trees, today's prompt	largest class	–	95.4%	1.2%	96.2%
Trees, signed prompt	largest class	–	96.9%	1.2%	94.7%
Trees, tables prompt	largest class	–	98.5%	1.2%	96.2%
Trees, today's prompt	threshold	99.85%	100%	13.5%	96.5%
Trees, signed prompt	threshold	99.76%	100%	6.3%	95.9%
Trees, tables prompt	threshold	99.9%	100%	3.0%	97.5%
GNN, residual-only	threshold	99.96%	100%	0.9%	96.5%

Source: S2, S1 (section 6). Thresholds reach the screen's calibration recall (99.3%): 0.003 to 0.008 for the LLM's probability. Left out: the diverged run (signed, 3 passes; AUC 28.8%).

Appendix: LLM Fine-Tunes

Run	Validation loss per answer token, every 128 steps	Scored
Today's, 1 pass	0.0356, 0.0335, 0.0336, 0.0361 (step 509)	step 256
Signed, 1 pass	0.0362, 0.0325, 0.0456, 0.0232 (step 509)	step 509
Tables, 1 pass	0.0549, 0.0331, 0.0212, 0.0201 (step 509)	step 509
Today's, 3 passes	0.0331 . . . 0.0260 (step 1,527); lowest 0.0241	step 1,408
Signed, 3 passes	**2.9063** at step 128, then 0.0765 . . . 0.0482	diverged
Tables, 3 passes, half rate	between 0.023 and 0.038 throughout	step 640

- 2.4 GPU hours per pass on an RTX PRO 6000. An answer is 19 to 21 tokens and one of them carries the decision: 0.033 per token is about 0.65 nats per decision.
- The signed three-pass run's training loss jumped above 8 at steps 82–86, near the peak learning rate. Its adapter requests on none of the 130 roots that need phasors.
- The half-rate run (5e-5) trained without a spike. Preempted runs resumed from their newest checkpoints.

Source: S1 (section 6), S4.

Appendix: Sources

S1 docs/classifier_triage_plan_20261004.md: plan, decisions, stage 0 to stage 2, contract changes (sections 1–12).

S2 output/classifier_triage_20261004/ieee14/: report.md, benchmark.json, probes, transfer_look_ieee57.md.

S3 output/classifier_triage_20261004/ieee14_location/: held-out location study.

S4 output/classifier_triage_20261004/llm, llm_results/; jobs in research/hpc/classifier_triage_20261004/.

S5 research/classifier_triage/ (features.py, model.py, runtime.py); export psse_env/oracle/models/triage_gnn_ieee14_20261004.

S6 output/classifier_triage_20261004/e2e/: expert end-to-end runs; _v2 runs after the trimmed contract.

S7 Cluster cell research_full_pipeline_20261007_classifier: out/r1/round_summary.json, round-2 receipts.

S8 Cluster cell research_full_pipeline_20261001_ranked: out/pipeline_summary.json.

S9 docs/slides/weekly_update_20261002.tex and output/hypothesis_ranking_20260930/screen_cost/: screen cost.

S10 psse_env/evidence_profile.py, actions.py, transactional_env.py, oracle/diagnostics_expert.py (commits 18c15f7, 11b7ad5).